Slant column density

What do we actually measure, and what is in it besides the atmosphere?

`Notebook companion: Unit-06-slant-columns.ipynb`

1 The question

Unit 4 established which wavelengths separate the species. Unit 5 established where along the line of sight the measurement is made. This unit puts the two together and asks what quantity the instrument actually delivers.

The answer matters more than it might appear, because the project leads with it. We report slant column density as the primary product and treat the vertical profile and the temperature as derived quantities. That decision depends on a claim about what the measurement contains, and this unit is where the claim is checked rather than asserted.

2 Physical picture

The detector records a signal in each row of each channel. Two of those signals matter. One is taken while the line of sight passes through the atmosphere. The other, the reference, is taken before the occultation begins. Their ratio is the transmission, and its logarithm is the optical depth:

$$T = \frac{I}{I_0}, \qquad \tau = -\ln T.$$

Almost everything that could go wrong with an absolute measurement cancels in that ratio. The detector's quantum efficiency cancels. The solar irradiance cancels, provided it has not changed between the two frames. The filter's transmission cancels. What does not cancel is the change in the relative response of the detector between the two moments, and Unit 11 is where the resulting requirement is addressed.

What survives the ratio is the atmosphere, the cross sections and the geometry. There is no temperature profile in it, no assumption of hydrostatic balance, and no assumption of diffusive equilibrium. That is the property we are claiming, and sections 4c and 4d test how far it holds.

3 Derivation

The measurement and its error

Because $\tau = -\ln T$, the uncertainty on the optical depth is the *fractional* uncertainty on the transmission: $\sigma_\tau = \sigma_T / T$. Two sources contribute.

Photon noise falls as the signal falls, so its contribution to σ_τ grows as $e^{\tau/2}$; the simulator carries it as $\sigma_\tau = 0.004\, e^{\tau/2}$ per 10 km bin.

The second source does not shrink with more photons. The ratio is taken between two frames separated in time, so it inherits any drift in the relative response of the pixels involved. If each frame carries a relative uncertainty ϵ from that drift, the ratio carries $\sqrt{2}\,\epsilon$, and this appears in σ_τ as a constant floor. For the one per cent pixel stability the vendor request for information asks about, the floor is 0.014.

A constant floor on σ_τ sets a lowest usable optical depth. The quantity that matters is the fractional error σ_τ/τ, and with σ_τ fixed that fraction grows without limit as τ falls. The same reasoning at the other end sets a highest usable optical depth, and the two together are what section 4a calls the window.

From optical depths to columns

Each channel's optical depth is a sum over species. In the optically thin limit, using the effective cross sections of Unit 4,

$$\tau_c \;=\; \sum_{s} \sigma^{\mathrm{eff}}_{cs}\, N_s ,$$

so with as many channels as species the columns follow by inverting a small matrix whose entries are cross sections and nothing else. No atmosphere enters that inversion. This is the whole argument for the slant column as the primary product, and section 4c shows both that the argument is sound and that this particular equation is not the one to use.

4 Worked numbers

4a. Where the optical-depth window comes from

τ	photon noise	systematic floor	total	fractional error
0.05	0.0041	0.0141	0.0147	29.4 %
0.10	0.0042	0.0141	0.0148	14.8 %
0.20	0.0044	0.0141	0.0148	7.4 %
1.00	0.0066	0.0141	0.0156	1.6 %
4.00	0.0296	0.0141	0.0328	0.8 %
8.00	0.2184	0.0141	0.2189	2.7 %

The lower end of the window is set by the detector, not by the photons. With a one per cent flat-field stability the measurement is good to ten per cent from $\tau = 0.15$ and to five per cent from $\tau = 0.30$. Photon noise on its own would allow τ down to 0.04. Whenever someone asks why the second question in the vendor request for information is about pixel-to-pixel stability rather than about quantum efficiency, this row is the answer.

The upper end of the window is not set by photon noise either. On photon noise alone the measurement stays inside ten per cent up to $\tau = 11$. What limits it first is out-of-band leakage. If a fraction f of the flux reaches the detector regardless of the atmosphere, the measured transmission cannot fall below f and the measured optical depth saturates at $-\ln f$. The bias appears well before that saturation point: a leak of 10^{-3} produces a five per cent underestimate from $\tau = 5.4$, and a leak

of 10^{-2} produces one from $\tau = 0.05$ — which is to say it corrupts the entire range rather than the deep end. The window $0.2 \le \tau \le 4$ that the project uses is consistent with a leak of a few parts in a thousand, and it is a specification on the filters rather than a property of the atmosphere.

Band saturation is a third effect and it is the mildest of the three. As the column grows, the wavelengths inside the passband that absorb most strongly go opaque first, and the channel's response flattens. Measured as $d\tau/d\ln N$ for the aluminium channel, it is 1.23 at $\tau = 1.43$ where the thin limit would give 1.43, and 2.69 at $\tau = 4.08$ where the thin limit would give 4.08. The channel's optical depth still rises with the column at the top of the window, but less than proportionally. Band saturation on its own would not have set the upper end of the window at four.

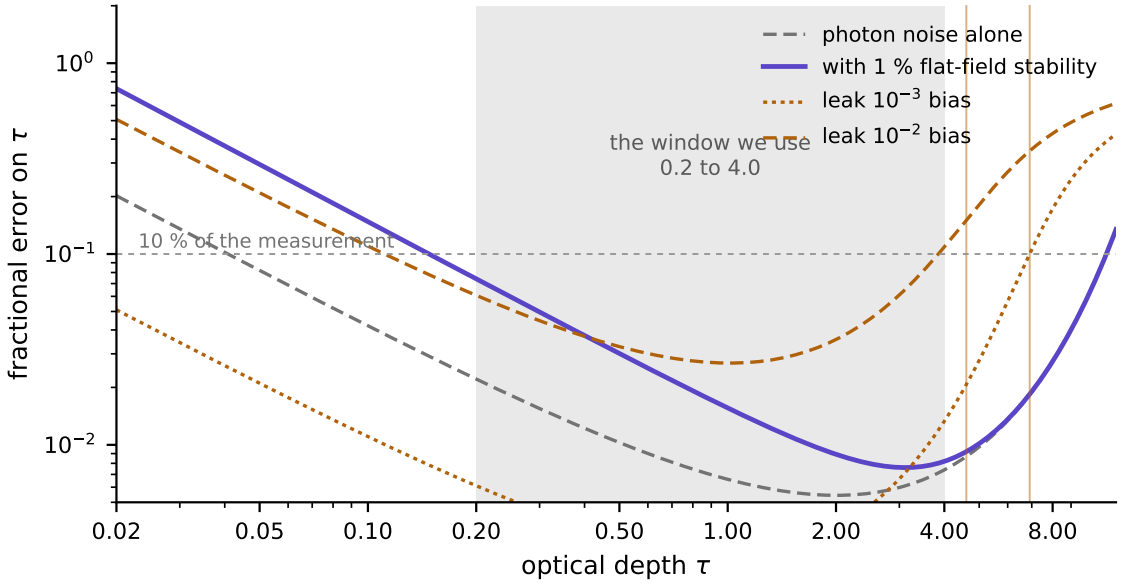


Figure 1: The two limits of the usable window. The solid curve is the fractional error on the optical depth with one per cent flat-field stability; the dashed grey curve is photon noise alone. The rising branch on the left is the systematic floor divided by a shrinking τ. The two brown curves are the fractional *bias* from an out-of-band leak of 10^{-3} and 10^{-2}, with the vertical lines marking where each leak makes the optical depth saturate. Generated by section D of the notebook.

4b. The altitude ladder

Applying that window to each channel gives the set of altitude bands, one per channel, that the team brief tabulates and that the rest of this series calls the ladder. Nothing in the table is chosen. It follows from the passbands of Unit 4 and the geometry of Unit 5.

Channel	Band /nm	Quiet	Storm, ap = 400
Zirconium	6–20	166–321 km	183–408 km
Aluminium	17–80	227–371 km	268–480 km
Indium	70–100	168–310 km	176–385 km
FUV	135–155	141–194 km	144–209 km
FUV	155–175	114–155 km	114–161 km

Two features of those bands matter later. The first is that they are staggered rather than stacked, so at most three of the five channels are ever inside the window at the same tangent altitude. The second is that a storm raises every band by roughly fifty kilometres, because the atmosphere expands and a given optical depth is then reached at a greater altitude.

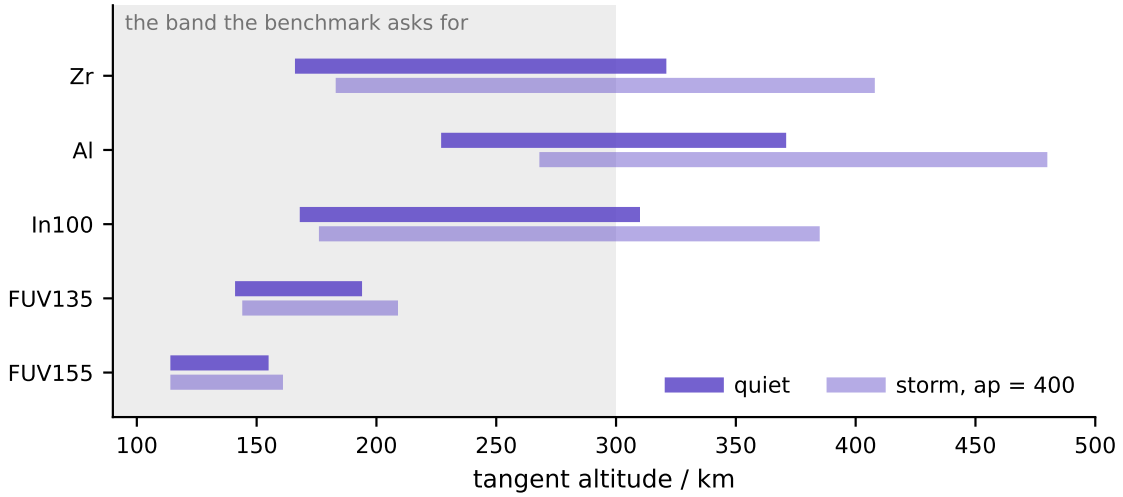


Figure 2: The altitude range over which each channel's optical depth lies inside the usable window, for a quiet and a severely disturbed atmosphere, with the band the benchmark asks for shaded. Generated by section E of the notebook.

4c. Inverting one tangent altitude

At 250 km the three extreme-ultraviolet channels are in window, with optical depths of 0.92, 2.50 and 0.70. Three equations, three species. Inverting the thin-limit matrix gives:

	true	linear	error	exact	error
O	2.572×10^{17}	2.795×10^{17}	+8.6 %	2.572×10^{17}	0.0 %
N_2	4.255×10^{16}	-1.917×10^{16}	−145 %	4.255×10^{16}	0.0 %
O_2	1.525×10^{15}	1.924×10^{16}	+1162 %	1.525×10^{15}	0.0 %

Columns in cm^{-2}. "Linear" inverts the thin-limit matrix of effective cross sections; "exact" inverts the band-averaged forward model numerically.

The linear inversion returns a negative nitrogen column. The reason is Unit 4's curve of growth: at optical depths near one the band average has already departed from $\sigma^{\mathrm{eff}}N$, the departure differs from channel to channel, and inverting a matrix built on the thin limit propagates those differences straight into the answer. Inverting the band-averaged forward model numerically recovers the truth exactly.

This does not weaken the claim that the product is model-free. The exact inversion uses the same three ingredients as the linear one — cross sections, the solar spectrum and the filter response —

and no atmosphere. It just cannot be written as a matrix inverse. The clean equation of section 3 is the right way to think and the wrong way to compute.

The conditioning of the inversion is unchanged by the switch to the exact form, and it is the quantity that justifies including indium:

Channels	**Solving for**	**Condition number**
Zr + Al	O, N_2	21.8
Zr + In	O, N_2	6.1
Al + In	O, N_2	7.5
Zr + Al + In	O, N_2	7.0
Zr + Al + In	O, N_2, O_2	29.6

A condition number of 21.8 means the measurement errors can be amplified twenty-two-fold on the way to the answer. Adding indium reduces that factor to 6.1. This is the same finding as Unit 4's contrast table, now expressed in the quantity that governs error amplification. It is the quantitative form of the statement that indium improves the conditioning of the inversion rather than the precision of any single measurement.

4d. What is determined, and where

A channel outside the window contributes nothing. A species that accounts for almost none of a channel's optical depth cannot be recovered from that channel, and Unit 4 measured what happens if it is left free anyway: the errors on the species that could be determined increase. So the honest product is not "the slant columns". It is, at each tangent altitude, the set of species that the channels inside the window actually constrain, reported with their errors. The rule is explicit and it is in the notebook:

1. Use the channels whose optical depth lies inside the window.
2. Free a species that carries more than two per cent of some usable channel's optical depth.
3. Drop from the free set, worst first, any species the measurement cannot determine to better than fifty per cent, and solve again.

z_t	**Channels in window**	**Solved for**	O	N_2	O_2
140	FUV155	O_2	—	—	3.8 %
170	Zr + In + FUV135	O, N_2, O_2	2.0 %	2.3 %	2.3 %
190	Zr + In + FUV135	O, N_2, O_2	2.2 %	3.4 %	6.0 %
220	Zr + In	O, N_2	2.1 %	3.8 %	—
250	Zr + Al + In	O, N_2	2.4 %	7.4 %	—
300	Zr + Al + In	O, N_2	5.1 %	30.1 %	—
340	Al	O	4.3 %	—	—

One-sigma precision on each species slant column, quiet atmosphere, with $\sigma_\tau = 0.0148$. A dash means the rule did not solve for that species at that altitude.

Collecting the altitudes at which each species is determined to better than ten per cent gives the coverage the instrument actually delivers:

Species	Quiet	Storm, ap = 400
O	168–370 km	185–450 km
N_2	168–260 km	178–298 km
O_2	115–192 km	115–208 km

The benchmark asks for the major species over 90 to 300 km. Oxygen covers most of that band and extends well above it. Nitrogen covers about a third of it. Molecular oxygen covers a different third, at the bottom. And there is only one narrow band, 168 to 192 km, where all three are determined at once, because that is the only place where an extreme-ultraviolet channel and a far-ultraviolet channel are both in window. Below 115 km nothing is measured at all.

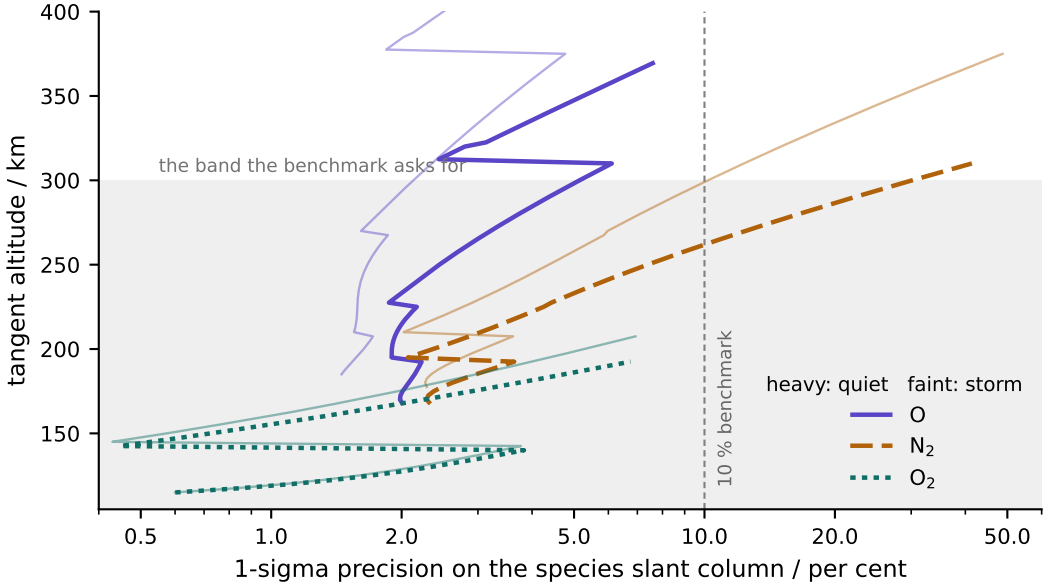


Figure 3: Precision on each species slant column against tangent altitude, quiet in heavy line and storm in faint. The steps are real: they occur where a channel enters or leaves the optical-depth window, or where the rule of section 4d starts or stops solving for a species. Generated by section G of the notebook.

4e. One column, with its whole error bar

At 180 km, with zirconium, indium and the 135–155 nm channel all in window:

Contribution	O	N_2	O_2
Photon noise	2.46 %	1.73 %	1.23 %
Flat-field stability, 1 %	1.98 %	2.50 %	3.56 %
I_0 residual, 500 km host at 10°	0.33 %	0.26 %	0.00 %
Combined precision	3.18 %	3.05 %	3.76 %
Accuracy, at 10 % cross-section error	9.1 %	9.1 %	9.1 %

The cross-section term does not enter through τ at all. It is an error in the matrix rather than in the measurement, and for a square system a fractional error in one species' cross section passes one-to-one into that species' column and leaves the others untouched. That is a consequence of linear algebra rather than of physics, and it stops being true the moment the system is overdetermined and solved by least squares.

5 What the computation shows

Both ends of the usable window are set by instrument specifications. The lower one is flat-field stability and the upper one is out-of-band rejection. Neither is a property of the atmosphere, and both are questions we ask vendors.

The product is model-free, but it is not linear. The matrix inverse of section 3 is the right way to understand what the measurement contains and the wrong way to compute it. Anyone reproducing our numbers from the thin-limit cross sections will get a negative nitrogen column at 250 km.

Coverage is a per-species statement. A single number for the altitude range is not available. Oxygen, nitrogen and molecular oxygen are each determined over a different band, they overlap in only 24 km, and the honest way to report the instrument is a table rather than a range.

6 What we decided

Slant column density remains the primary product, and section 4c is why: the inversion from measured optical depths to species columns uses cross sections, the solar spectrum and the filter response, and no atmospheric model. That property survives the failure of the linear form.

We report coverage per species, with the table of section 4d, rather than as a single altitude range. The claim that the instrument covers 90 to 300 km was never true of the retrieval and is now replaced by something we can defend.

The species-selection rule of section 4d becomes part of the product definition rather than an implementation detail. A retrieval that reports a number for a species its channels do not constrain is worse than one that reports nothing, and the threshold that separates the two cases has to be published.

The window $0.2 \le \tau \le 4$ is restated as two instrument requirements: pixel-to-pixel stability of one per cent or better over a year, and out-of-band rejection better than about 10^{-3}. Both go into the vendor questions.

7 What we do not claim

Not that the assimilation argument is stronger than it is. Earlier project documents say that density-model assimilation ingests observations with characterized errors "not other people's retrieved profiles". As a description of current practice that is wrong, and a reviewer in this community will know it: retrieved products are routinely assimilated, including GOLD's retrieved O/N_2 column ratio into WACCM-X [4] and COSMIC's retrieved electron density profiles into TIEGCM [5]. What is defensible is narrower and still supports the decision. Thermospheric assimilation systems ingest observations through an observation operator with a specified error covariance [5, 1]. Numerical weather prediction found that assimilating the measured quantity rather than a retrieved profile avoids double-counting the prior information built into the retrieval, and that the change improved forecast skill [2]; the two are equivalent only when the retrieval's prior does not underrepresent the variability of the state and its error covariance is propagated correctly [6]. Reporting a slant column with a documented error budget keeps both routes open for whoever uses it. Correct the older wording wherever it appears.

Not that the error budget here is complete. It contains photon noise, flat-field drift, the reference-frame residual and cross-section uncertainty. It does not contain pointing error, which enters as an error in tangent altitude rather than in optical depth; the horizontal structure the spherically symmetric geometry of Unit 5 ignores; or the temperature dependence of the cross sections.

Not that the precision figures are achievable at every altitude in the table. They assume $\sigma_\tau = 0.0148$ at every channel, which in turn assumes the one per cent flat-field stability has been demonstrated. It has not been. It is the second question in the vendor request for information and the subject of a Phase 1 bench measurement.

Not that the species-selection thresholds are principled. Two per cent and fifty per cent are round numbers chosen to make the rule behave sensibly. Exercise E6.3 asks what would justify them, and the honest answer today is that nothing does beyond the resulting coverage.

8 Exercises

E6.1 The lower end of the window moves with the flat-field stability. Recompute the usable optical-depth range for stabilities of 0.3, 1 and 3 per cent, and turn each into the altitude range the zirconium channel would cover. This is the quantitative version of the second question in the vendor request for information, and the answer tells you how much of the instrument's coverage depends on that one specification.

E6.2 Give one channel an out-of-band leak of 10^{-3} by replacing its transmission with $e^{-\tau} + f$, then rerun the section 4d solve at 250 km. Report the bias in each species column. Identify which species is biased most, and explain why it is that one in terms of the condition number rather than the size of the leak.

E6.3 The rule in section 4d drops a species when its precision is worse than fifty per cent. Vary that threshold between twenty and two hundred per cent and plot the resulting coverage for nitrogen. Then state which threshold you would defend to a reviewer, on grounds other than the coverage it produces.

References

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[3] J. A. Fennelly and D. G. Torr, "Photoionization and photoabsorption cross sections of O, N_2, O_2, and N for aeronomic calculations," *At. Data Nucl. Data Tables* **51**(2), 321–363 (1992).

[4] F. I. Laskar, N. M. Pedatella, M. V. Codrescu, R. W. Eastes and J. L. Anderson, "Assessing the impacts of assimilating GOLD disk O/N_2 observations on the thermosphere–ionosphere system," *J. Geophys. Res. Space Physics* **129**(12) (2024), doi:10.1029/2024JA033163.

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[6] S. Migliorini, "On the equivalence between radiance and retrieval assimilation," *Mon. Weather Rev.* **140**(1), 258–265 (2012), doi:10.1175/MWR-D-10-05047.1.

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[8] G. R. Smith and D. M. Hunten, "Study of planetary atmospheres by absorptive occultations," *Rev. Geophys.* **28**(2), 117–143 (1990), doi:10.1029/RG028i002p00117.

[9] E. M. B. Thiemann et al., "Neutral density profiles from PROBA2/LYRA solar occultations," *Space Weather* **15**, 1649 (2017); arXiv:1708.08833.

Every quantitative result in this unit is provisional in the four ways that apply to the whole project: filter bands are idealized top-hats rather than measured transmission curves; the truth atmosphere is NRLMSIS 2.1; helium is omitted above 400 km; and no horizontal structure along the ray is modelled. Analysis code is in `channel-plan/`.